



Week 3 Review: Probability and Distributions

The 142 GSI Team

Material Covered:

- Basics of Probability
 - Terms, Rules
 - Independent / Disjoint Events
- Distributions
 - Normal
 - Binomial
 - Poisson
- Sampling and the CLT

Introduction to Probability

- Definition: Probability describes how likely outcomes are to occur in random phenomena.
 - Look for keywords in probability questions: "given," "and," "or," "both," "neither."
- Another definition: Probability measures how frequently an outcome occurs if the random process is repeated many times.
 - More formally: $\text{Probability} = (\# \text{ of times an outcome occurs}) / (\text{total } \# \text{ of repetitions})$ as repetitions approach infinity.

Sample Space

Sample Space (**S**): All possible outcomes of a random process.

- Discrete Example: {Disease, No Disease}
- Continuous Example: Range of possible GPAs (e.g., 0.0 - 4.0)

Event: A subset of outcomes within the sample space (e.g., $\text{GPA} \geq 3.0$, having a disease).

Probability Model

- A well-defined probability **model** transforms uncertainty into measurable values, allowing us to reliably interpret outcomes and make informed decisions.
- A model consists of:
 - Sample Space (S): Clearly defined **set of possible outcomes**.
 - Probabilities: Assigned to outcomes/events based on how frequently they're expected to occur.
- Example: A Model for flipping a fair coin:
 - Sample Space: {Heads, Tails}
 - Probability of Heads = Probability of Tails = 0.5

Probability Rules

- **Rule 1:** Probabilities must be between 0 and 1:

$$0 \leq P(A) \leq 1$$

- **Rule 2:** All possible outcomes of a sample space (S) together have a probability of 1:

$$P(S) = 1$$

- **Rule 3:** If two events have a joint probability of 0 (i.e., no overlap in their event spaces) then they are disjoint and the probability of either event occurring is the summation of their individual probabilities.

- If A and B are disjoint: $P(A \cap B) = 0$
- If A and B are disjoint: $P(A \cup B) = P(A) + P(B)$

- **Rule 4:** The probability of an event not occurring is 1 minus the probability of the event occurring. This event is called the complement and is denoted by a superscripted c or a single quotation mark.

$$P(A') = P(A^c) = P(A \text{ does not occur}) = 1 - P(A)$$

Dependence and Independence

- Independent events
 - Two events A and B are independent if knowing that one event occurred does not change the probability that the other occurred
 - Multiplication Rule for Independent Events:
 - If A and B are independent:
$$P(A \cap B) = P(A) \times P(B)$$
 - Conversely, if this condition is not satisfied, then events A and B are dependent.
- Dependent events
 - Two events A and B are dependent if knowing that one event occurred **does** change the probability that the other occurred
 - Multiplication Rule for Dependent Events:
 - If A and B are dependent,
$$P(A \cap B) = \mathbf{P(A|B)} \times P(B)$$
$$= \mathbf{P(B|A)} \times P(A)$$
- **Disjoint** events
 - Two events are disjoint if they **cannot both occur at the same time.**
 - This means the events are **mutually exclusive.**
 - If A and B are disjoint, $P(A \cap B) = 0$

More on Independent Events

- Independent events cannot be disjoint
- On a Venn diagram, independent events must overlap
- If two events overlap, they may or may not be independent
- To check if two events are independent you can perform any of the following checks:
 - Does $P(A \text{ and } B) = P(A) * P(B)$? If it does, A and B are independent
 - Does $P(A|B)=P(A)$? If it does, then A and B are independent.
 - Does $P(B|A)=P(B)$? If it does, then A and B are independent.

More on Disjoint Events

- Two events are disjoint if they cannot both occur at the same time. This means the events are mutually exclusive
- Examples:
 - You get a test for COVID-19, you either test positive or test negative. You cannot test both positive or negative at the same time.
- **Disjoint events are always dependent!** This is because knowing that the one event occurs implies there is a 0% chance the other event will also occur.

Probability Calculation Rules for Two Events

- Conditional Probability
 - When $P(A) > 0$, the conditional probability of B given A is:
$$P(B|A) = P(A \cap B) / P(A)$$
- General addition rule
 - For any two events A and B:
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
 - This simplifies to:
$$P(A \cup B) = P(A) + P(B) \text{ when A and B are disjoint}$$
- General multiplication rule
 - This is simply a rearrangement of the conditional probability formula:
$$P(A \cap B) = P(A)P(B|A)$$
 - This simplifies to:
$$P(A \cap B) = P(A) \times P(B) \text{ when A and B are independent}$$
- Bayes' Theorem
 - $P(A|B) = P(A)$ if independent. But if they aren't:
$$P(A|B) = P(B|A)P(A) / [P(B|A)P(A) + P(B|A^c)P(A^c)]$$

Probability Example: Hearing Impairment in Dalmatians (3 min exercise)

Congenital sensorineural deafness is the most common form of deafness in dogs and is often associated with congenital pigmentation deficiencies. A study of hearing impairment in dogs examined over 5000 dalmatians for both hearing impairment and iris color. Being impaired was defined as deafness in either one or both ears. Dogs with one or both irises blue (a trait due to low iris pigmentation) were labeled blue. The study found that 28% of the dalmatians were hearing impaired, 11% were blue eyed, and 5% were hearing impaired and blue eyed.

Questions:

- 1. Use probability notation to write the known probabilities from the prompt.**
- 2. What is the probability of dalmatians being either blue eyed or hearing impaired?**
- 3. Are blue eyes and impaired hearing independent or dependent? Why?**

Solution

1. $P(I) = 0.28$
 $P(B) = 0.11$
 $P(B \cap I) = 0.05$
2. $P(B \cup I) = P(B) + P(I) - P(B \cap I) = 0.11 + 0.28 - 0.05 = 0.34$
3. $P(B) * P(I) = 0.28 \times 0.11 = 0.0308 \neq P(B \cap I) = 0.05$
So events B and I are dependent.

Diagnostic Testing

- **Sensitivity:** The test's ability to appropriately give a **positive** result when a person has the disease, or $P(\text{Test} = + | \text{Disease} = \text{TRUE})$
 - or, Sensitivity = $1 - \text{False Negative Rate}$
- **Specificity:** The test's ability to appropriately give a **negative** result when a person tested does not have the disease, or $P(\text{Test} = - | \text{Disease} = \text{FALSE})$
 - or, Specificity = $1 - \text{False Positive Rate}$
- **Positive predictive value:** The chance that a person truly has the disease, given that the test is positive, or $P(\text{Disease} = \text{TRUE} | \text{Test} = +)$
- **Negative predictive value:** The chance that a person truly does not have the disease, given that the test is negative, or $P(\text{Disease} = \text{FALSE} | \text{Test} = -)$

Predictive Values will always condition on the test result first!

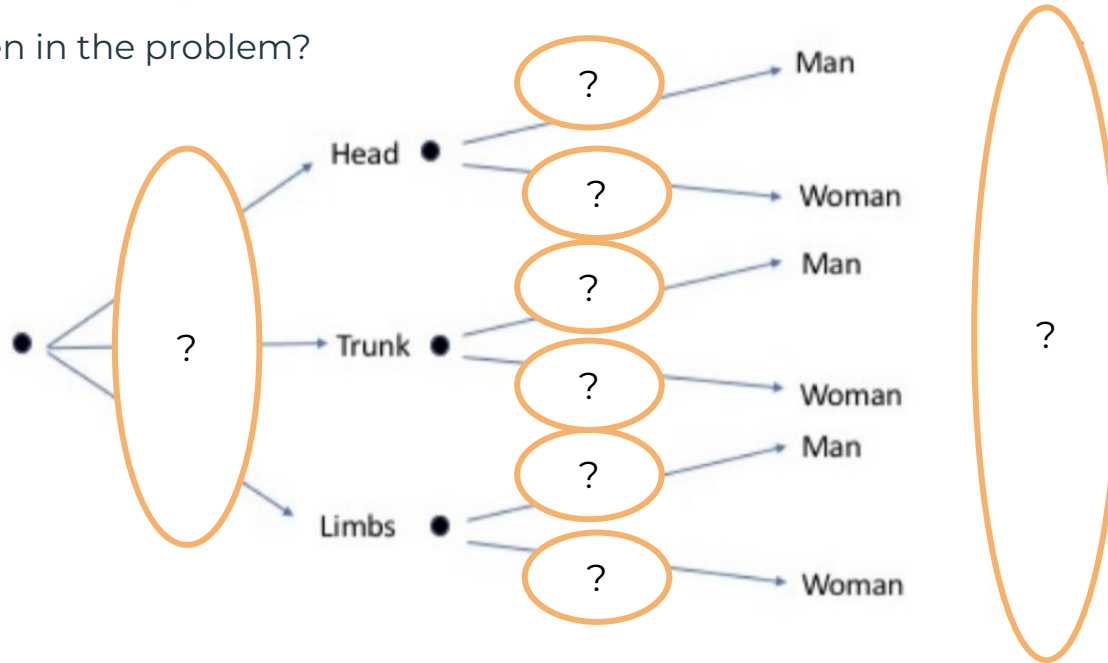
Example: Absolute Frequencies and Tree Diagrams (5 min exercise)

A study of the presence of cutaneous malignant melanoma among the Italian population found that 15% of skin cancers are located on the head and neck area, another 41% on the trunk and the remaining 44% on the limbs. 44% of individuals with skin cancer on the head are men; similarly, 63% of those with skin cancer on the trunk, and 20% of those with skin cancer on the limbs are men. **What percent of all individuals with skin cancer are women?**

Solution: Tree Diagram

$P(A)$ Event A $P(B|A)$ Event B $P(A \text{ and } B)$

What's given in the problem?



1. Check for Given Probabilities (marginal / conditional)

A study of the presence of cutaneous malignant melanoma among the Italian population found that **15%** of skin cancers are located on the **head** and neck area, another **41%** on the **trunk** and the remaining **44%** on the **limbs**.

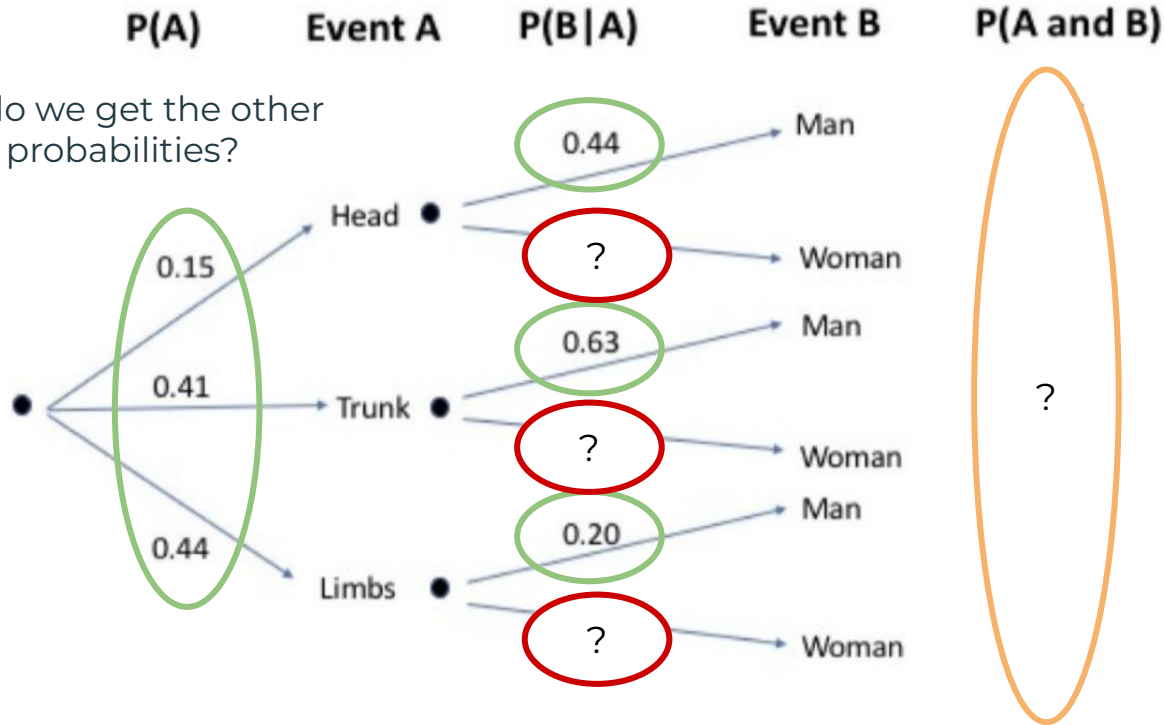
(Marginal probabilities)

44% of individuals with skin cancer on the **head** are **men**; similarly, **63%** of those with skin cancer on the **trunk**, and **20%** of those with skin cancer on the **limbs** are **men**.

(Conditional probabilities)

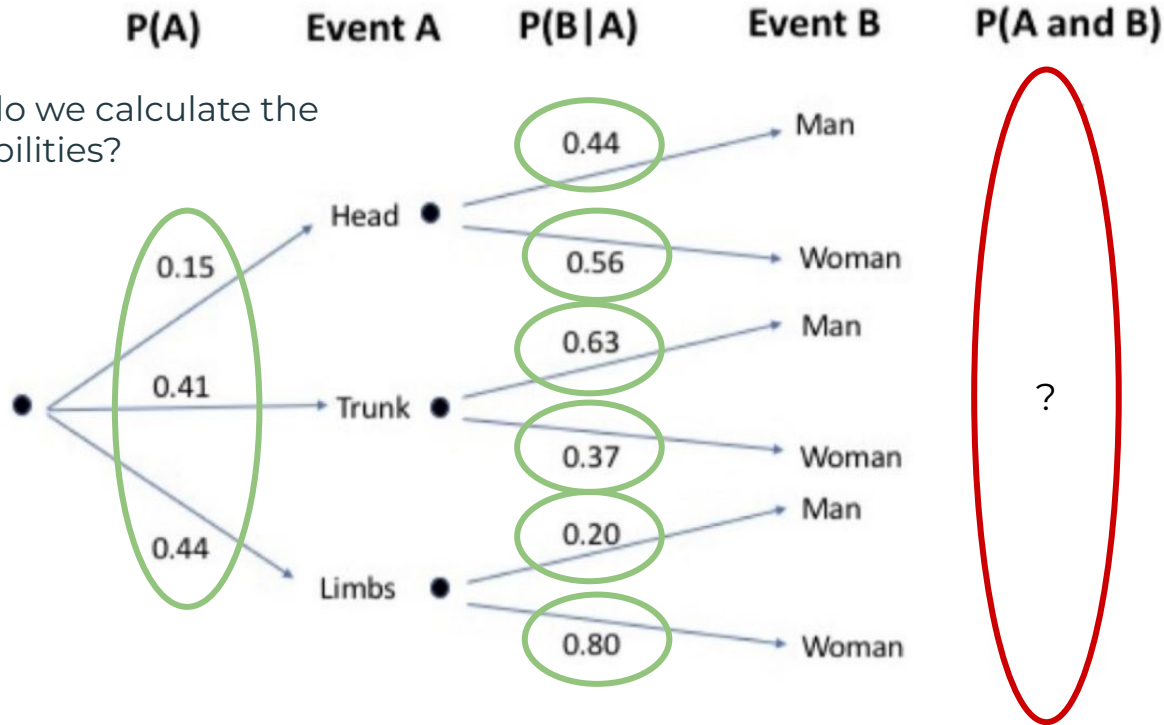
2. Calculate Complementary Conditional Probabilities

Now, how do we get the other conditional probabilities?



3. Calculate Joint Probabilities

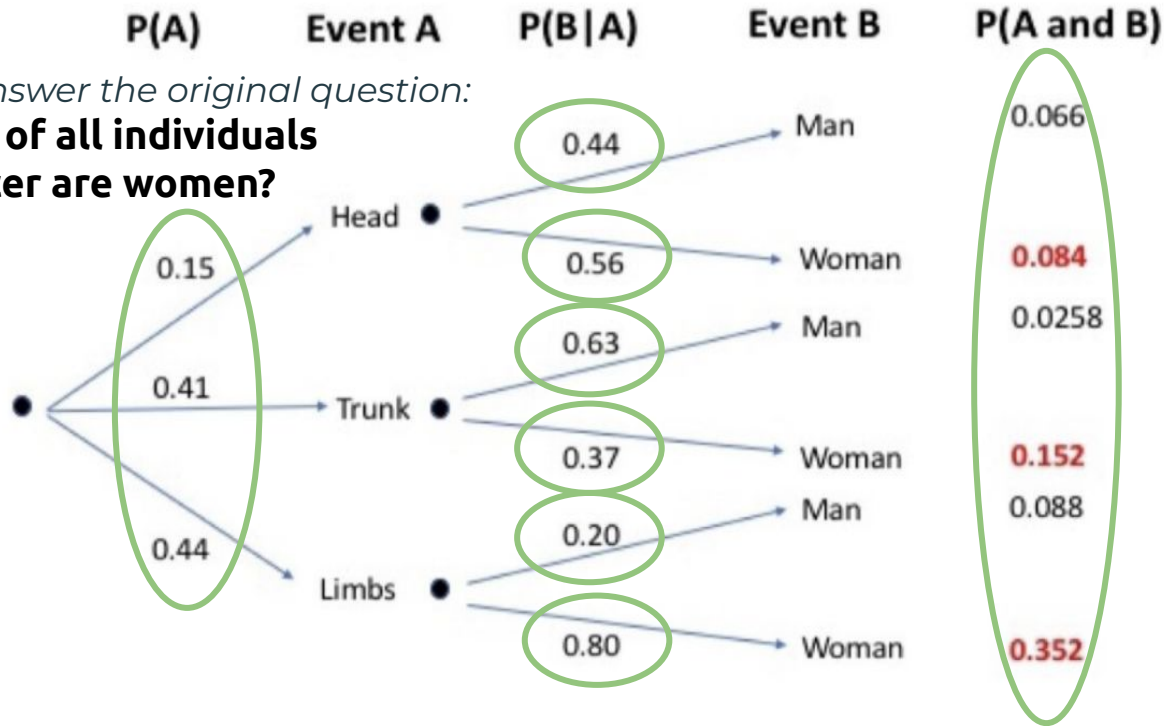
Now, how do we calculate the joint probabilities?



4. Calculate the (Trickier) Marginal Probability

We can now answer the original question:

What percent of all individuals with skin cancer are women?



$$P(A) \times P(B|A) = P(A \text{ and } B),$$

i.e. multiply across the tree probabilities

Solution

$P(\text{head}) = 0.15$
 $P(\text{trunk}) = 0.41$
 $P(\text{limbs}) = 0.44$

Marginal
Probabilities

$P(\text{man} \mid \text{head}) = 0.44$
 $P(\text{man} \mid \text{trunk}) = 0.63$
 $P(\text{man} \mid \text{limbs}) = 0.20$

Conditional
Probabilities

$P(\text{woman} \mid \text{head}) = 1 - 0.44 = 0.56$
 $P(\text{woman} \mid \text{trunk}) = 1 - 0.63 = 0.37$
 $P(\text{woman} \mid \text{limbs}) = 1 - 0.2 = 0.8$

$$\begin{aligned} \mathbf{P(\text{woman})} &= P(\text{woman} \ \& \ \text{head}) + P(\text{woman} \ \& \ \text{trunk}) + P(\text{woman} \ \& \ \text{limbs}) \\ &= P(\text{head})P(\text{woman}|\text{head}) + P(\text{trunk})P(\text{woman}|\text{trunk}) + P(\text{limbs})P(\text{woman}|\text{limbs}) \\ &= (0.15 * 0.56) + (0.41 * 0.37) + (0.44 * 0.8) = \mathbf{0.585} \end{aligned}$$

Two by Two Tables: Risk and Odds

	Has Disease (D+)	No Disease (D-)	Total
Is Exposed (E+)	100	50	150
Not Exposed (E-)	50	300	350
Total	150	350	500

- What is the **risk (a.k.a. probability)** that someone has the disease?
 - Risk often concerns a **marginal** probability: $P(\text{Disease})$
 $P(\text{Has disease}) = P(D+) = 150/500 = 0.3 = 30\%$ (call this probability “p” for now).
 - Risk of having the disease in this population = 30%
- What is the **odds** that someone has the disease?
 - **Odds of having the disease:** $p/(1-p) = (150/500)/(350/500) = 150/350 = 3/7$
 - Alternate calculation: Odds of having the disease: $p/(1-p) = 0.3/0.7 = 3/7$ “3 to 7 odds of having the disease)

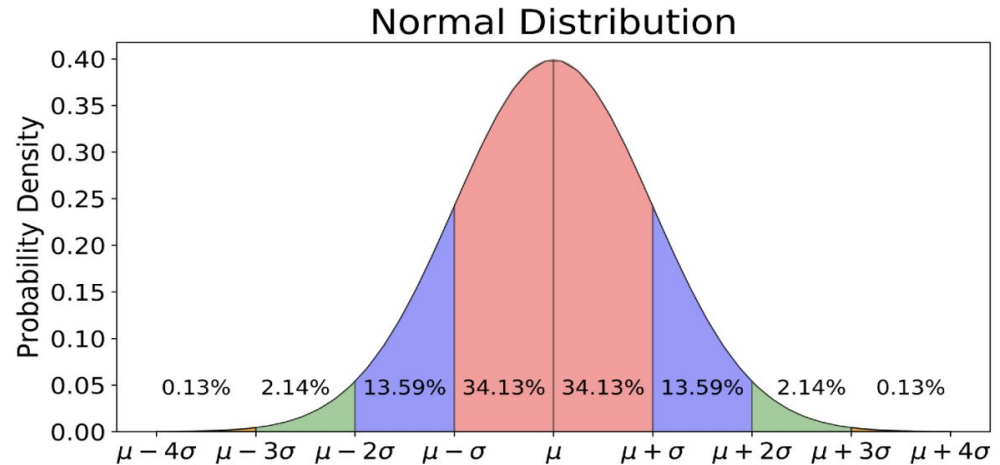
Distributions

- Use distributions to model patterns in data
- For instance:
 - **Normal, Binomial, Poisson**
- We can calculate the distribution's mean, variance/standard deviation, and calculate probabilities based on what we know about the data's underlying distribution

Normal Distribution (Continuous)

We want to find a probability for greater than or less than or greater/equal, or less/equal, or falling within a certain interval, but **NEVER** exactly equal to a single value

$$\text{If } X \sim N(\mu, \sigma) \rightarrow P(X = x) = 0$$



Normal Distribution Functions

Function	Use	Example
<code>pnorm()</code>	Outputs a probability , input is a quantile value. Compute the probability at a specified x or below . Is a cumulative probability. Must specify the mean and sd. Remember $P(X=x) = 0 \rightarrow$ i.e., $P(X<x) = P(X\leq x)$	$P(X<2)$: <code>pnorm(2, mean = 3, sd = 0.5)</code> $P(X>2)$: <code>pnorm(2, mean = 3, sd = 0.5, lower.tail = F)</code>
<code>rnorm()</code>	Generate random draws from a normal distribution. We use this to create/simulate data to get a sense of what normally distributed data looks like.	<code>rnorm(10, mean = 3, sd = 0.5)</code> generates 10 random draws from a Normal distribution
<code>qnorm()</code>	Outputs a quantile , input is a probability. Works as the inverse of <code>pnorm()</code> . In other words, provide a percentage and this returns the x value such that x% of the area is at/below that value in the lower tail.	$P(X\leq x) = 75\%$, what is x? <code>qnorm(0.75, mean = 3, sd = 0.5)</code> will return the value x.

Normal Distribution

- Remember for the Normal distribution: $P(X = x) = 0$
 - ex: $P(X = 2) = 0$
 - This property is true for any continuous distribution – the only one we study is the Normal distribution.
- This is NOT true for Binomial and Poisson; they can have non-zero probabilities of being exactly equal to whole numbers (like 2).
 - For Binomial or Poisson, $P(X < 2) = P(X = 0) + P(X = 1)$,
but $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$

Z - Value

- ▶ Any random variable that follows a Normal distribution can be standardized
- ▶ If x is an observation from a distribution that has a mean μ and a standard deviation σ ,

$$z = \frac{x - \mu}{\sigma}$$

- ▶ Interpretation: z is the number of standard deviations that x is above or below the mean of the data.

QQ Plot: Checking for Normality

Use a histogram and a qq-plot to test whether your data is normal. In our `birth.weight` data, the variable is named `weight`.

```
# check data distribution
```

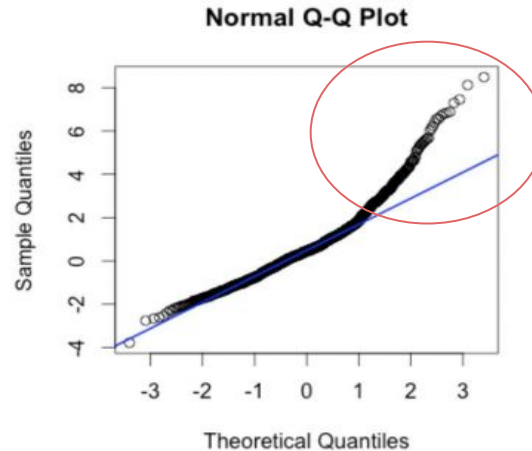
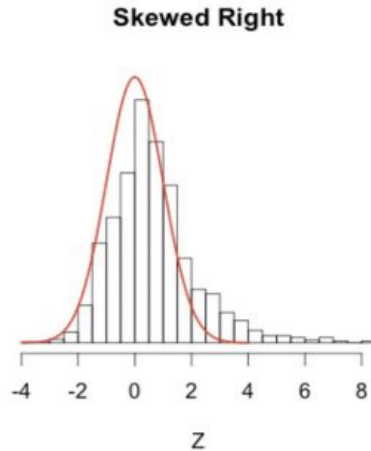
```
plot.hist <- ggplot(data = birth.weights, aes(x=weight)) +  
  geom_histogram()
```

```
# check quantile plot
```

```
plot.qq <- ggplot(data = birth.weights, aes(sample=weight)) +  
  stat_qq() +  
  stat_qq_line()
```

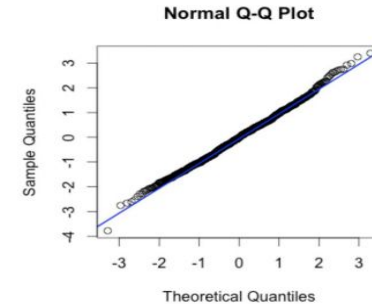
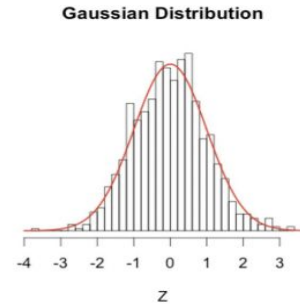
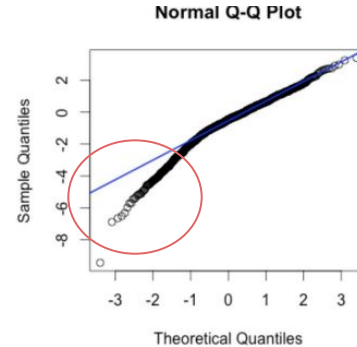
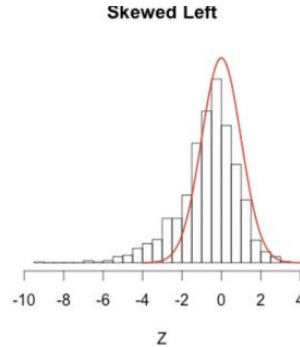
Understanding Skew Using QQ Plots

Skewed right data deviates **positively** towards the right top corner. If data were normally distributed, 99.7% of the data should theoretically be kept between -3 and 3, but here the sample quantiles are between 4 and 8 -> way outside of the range for normal data for the data on the right-hand side of the histogram.



Understanding Skew Using QQ Plots

Similarly, **skewed left** data deviates **negatively** towards the bottom left corner.



And normal data falls on the straight line.

Binomial Distribution (Discrete)

- Distribution for discrete random variables (only whole numbers)
- Written as: $X \sim \text{Bin}(n,p)$
 - X : the random variable counting the number of “successes” out of n trials
 - n : total number of “trials”
 - p : probability of success
- We can start approximating the Normal distribution using the binomial when n is large.
 - General rule: Use when $n \cdot p \geq 10$ and $n \cdot (1-p) \geq 10$

Binomial Assumptions

Must meet these assumptions:

1. You have some fixed amount of trials
2. Trials are independent
3. The probability of success is the same for each trial
4. There are only two outcomes: success or failure

Binomial Calculation

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Here's what each term represents:

- $\binom{n}{k}$ is the binomial coefficient, also known as "n choose k," which calculates how many ways k successes can be achieved in n trials.
- p^k is the probability of getting k successes.
- $(1 - p)^{n-k}$ is the probability of getting $n - k$ failures.

Calculating the combination function

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

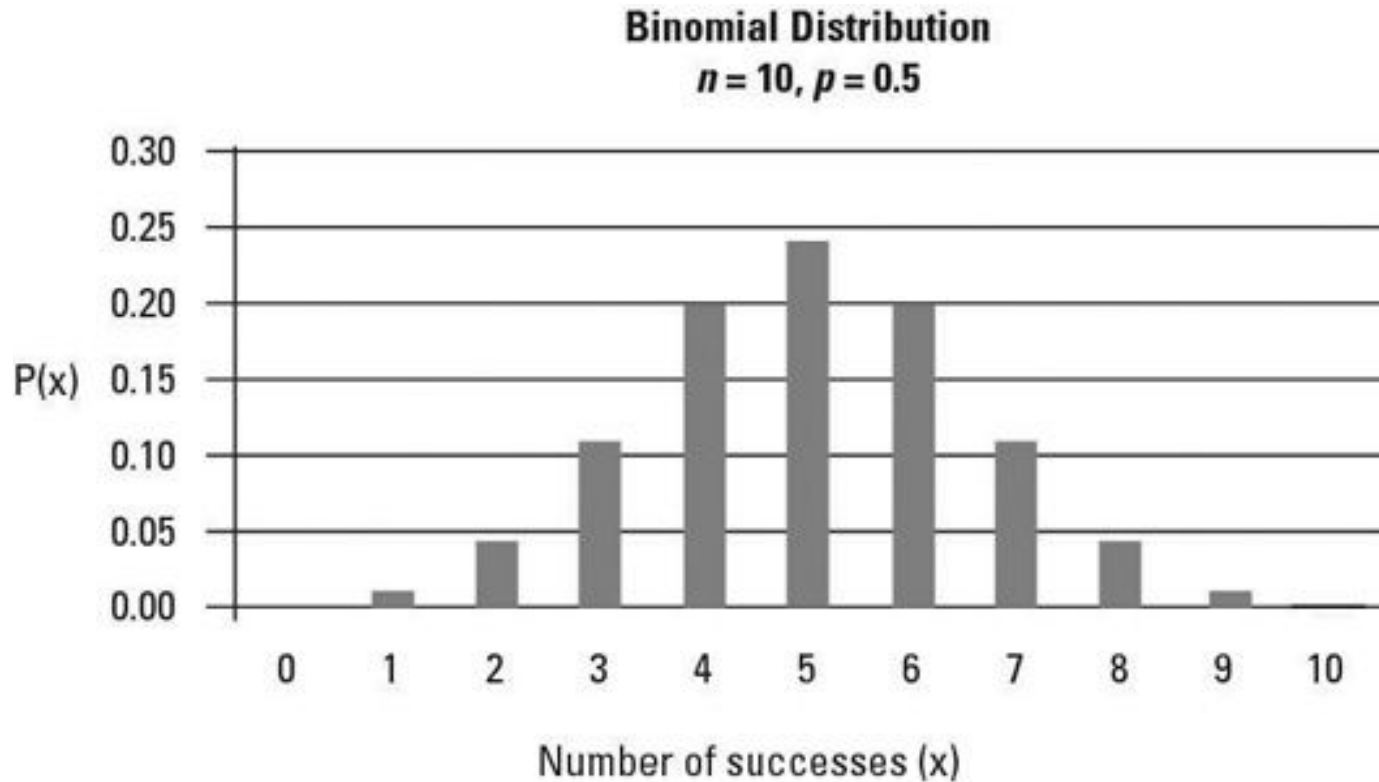
where $n!$ denotes the factorial of n , which is the product of all positive integers up to n .

Steps to Calculate by Hand:

Example: Calculate $\binom{5}{2}$

- Calculate $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
- Calculate $2! = 2 \times 1 = 2$
- Calculate $(5 - 2)! = 3! = 3 \times 2 \times 1 = 6$
- Calculate $\binom{5}{2} = \frac{120}{2 \times 6} = \frac{120}{12} = 10$

Binomial Distribution



Binomial Functions

Function	Use	Example
<code>pbinom()</code>	calculate the cumulative probabilities (goes from left to right, i.e. 0 up to the chosen value) i.e. $P(X \leq 2)$	If $X \sim \text{Binomial}(10, 0.5)$, to find $P(X \leq 2)$: <code>pbinom(2, 10, 0.5)</code>
<code>dbinom()</code>	calculate the probability mass function, i.e. $P(X = 2)$	If $X \sim \text{Binomial}(10, 0.5)$, to find $P(X=2)$: <code>dbinom(2, 10, 0.5)</code>
<code>rbinom()</code>	generate random draws from a binomial distribution	To generate 3 random numbers from a $\text{Binomial}(10, 0.5)$: <code>rbinom(3, 10, 0.5)</code>

Normal Approximation for the Binomial Distribution

- If we have a binomial distribution that satisfies the below properties, we can approximate the distribution as a normal distribution
- This allows us to use all the useful properties from normal distributions (very convenient!)

NORMAL APPROXIMATION FOR BINOMIAL DISTRIBUTIONS

Suppose that a count X has the binomial distribution with n observations and success probability p . When n is large, the distribution of X is approximately Normal, $N(np, \sqrt{np(1-p)})$.

As a rule of thumb, we will use the Normal approximation when n is so large that $np \geq 10$ and $n(1-p) \geq 10$.

Binomial Example (5 min exercise)

The probability that a patient recovers from a widespread flu is 90%. There are now 30 people in a hospital who have contracted the flu independently.

1. What is the probability that 25 patients in that hospital will recover from it?
2. How about more than 25 patients?

Solution (Q1)

The prompt information translates to: $X \sim \text{Bin}(30, 0.9)$, for Q1 we are finding $P(X = 25)$, i.e.

```
n_trials <- 30  
k_success <- 25  
probability <- 0.9
```

And we can calculate the first quantity “by hand” using the formula:

```
choose(n_trials, k_success)*(probability)^k_success*(1-probability)^(n_trials-k_success)  
[1] 0.1023048
```

Or we can use the ``dbinom()`` function in R:

```
dbinom(x = k_success, size = n_trials, prob = probability)
```

Solution (Q2)

The second quantity, $P(X > 25)$, can be found using the `pbinom` function that first calculates the sum of probabilities from 0 to k and then takes the complement:

```
1-pbinom(q=k_success, size=n_trials, prob=probability)
[1] 0.8245051
```

Or even more conveniently:

```
pbinom(q=k_success, size=n_trials, prob=probability, lower.tail=FALSE)
[1] 0.8245051
```

Poisson Distribution

The Poisson distribution is generally used to model the count of **rare events** occurring within a **fixed interval** (time, space, or other measure) when events happen independently and at a constant average rate:

- Examples: number of phone calls received at a call center per hour, number of raindrops hitting a 1 m² area per minute, number of arrivals at an emergency department within a 3-hour period

The formula for the Poisson distribution with mean/rate parameter lambda:

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Must meet these assumptions:

1. Events occur independently.
2. The rate at which events occur is constant, i.e, the rate cannot be higher in some intervals and lower in other intervals.
3. No upper limit (like Binomial)

Poisson Functions (Similar to Binomial)

Function	Use	Example
<code>ppois()</code>	Compute the probability at a specified x or below (lower.tail=T by default). Is a cumulative probability. Need to also specify lambda, the mean of the Poisson distribution	$P(X \leq 2)$: <code>ppois(2, lambda = 0.5)</code> $P(X < 5)$: <code>ppois(4, lambda = 0.5)</code> $P(X > 5) = 1 - P(X \leq 5) = 1 - \text{ppois}(5, \text{lambda} = 0.5)$
<code>dpois()</code>	Compute an exact probability at a specified value for x, where x is a discrete value between 0 and infinity. Need to also specify lambda, the mean of the Poisson distribution	$P(X=2)$: <code>dpois(2, lambda = 0.5)</code> $P(X=1) + P(X=2)$ (can add them together)
<code>rpois()</code>	Generate a random variable that is drawn from a Poisson distribution. We use this to create/simulate data to get a sense of what Poisson data looks like	<code>rpois(10, lambda = 0.5)</code> generates 10 random draws from a Poisson distribution